THE SCHEME THEORETIC IMAGE OF A QUASI-COMPACT MORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author refers back to a computation performed in the proof of the scheme theoretic image lemma for assertion (2); that reference is to a statement in the same chapter and the counted proof supplies the quasi-coherence argument for I in full.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The scheme theoretic image of a quasi-compact morphism and its defining ideal. Lemma lemma-scheme-theoretic-image, used at proof lines 3268-3271. Supplies the identification of Z with the closed subspace defined by I once (1) holds.
- The pushforward of the structure sheaf along a quasi-compact quasi-separated morphism is quasi-coherent. Lemma lemma-pushforward, used at proof lines 3276-3280. Makes f'_*O_U quasi-coherent, so that its kernel is.
- Affine etale covers of quasi-compact morphisms and etale locality of quasi-coherence. used at proof lines 3272-3275. Reduces quasi-coherence of I to the case of an affine base.
- Density of the image in the scheme theoretic image. used at proof lines 3281-3283. Derives assertion (4) from the description of I.

Source reference. Original lemma 03M9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4741–4741. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 082X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10267–10267. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $Z \subset Y$ be the scheme theoretic image of f. If f is quasi-compact then*

(1) *the sheaf of ideals $\mathcal{I} = \mathrm{Ker}(\mathcal{O}_Y \to f_*\mathcal{O}_X)$ is quasi-coherent,*
(2) *the scheme theoretic image Z is the closed subspace corresponding to $\mathcal{I}$,*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(3) *for any étale morphism $V \to Y$ the scheme theoretic image of $X \times_Y V \to V$ is equal to $Z \times_Y V$, and*
(4) *the image $|f|(|X|) \subset |Z|$ is a dense subset of $|Z|$.*

Proof. To prove (3) it suffices to prove (1) and (2) since the formation of $\mathcal{I}$ commutes with étale localization. If (1) holds then in the proof of Lemma 082X we showed (2). Let us prove that $\mathcal{I}$ is quasi-coherent. Since the property of being quasi-coherent is étale local we may assume Y is an affine scheme. As f is quasi-compact, we can find an affine scheme U and a surjective étale morphism $U \to X$. Denote f' the composition $U \to X \to Y$. Then $f_*\mathcal{O}_X$ is a subsheaf of $f'_*\mathcal{O}_U$, and hence $\mathcal{I} = \mathrm{Ker}(\mathcal{O}_Y \to \mathcal{O}_{X'})$. By Lemma 03M9 the sheaf $f'_*\mathcal{O}_U$ is quasi-coherent on Y. Hence $\mathcal{I}$ is quasi-coherent as a kernel of a map between coherent modules. Finally, part (4) follows from parts (1), (2), and (3) as the ideal $\mathcal{I}$ will be the unit ideal in any point of $|Y|$ which is not contained in the closure of $|f|(|X|)$. □